Online appendix for "Endogenous Plucking Through Networks: The Plucking Paradox"

Contents

A Derivations for the static equilibrium **2**

B Alternative calibration without targeting output-growth skewness **5**

B.1 Parameter values 5

B.2 Calibration targets and fit 6

B.3 Endogenous-variable means 7

C Simulation Exercises **7**

C.1 Ergodic moments 7

C.2 Joint size and duration dependence 8

C.3 Fixed SDF 8

D A plucking-cycle exercise in simulated output **9**

E Additional empirical results **11**

F Canonical RBC model for benchmark **12**

G CES extensions for unit costs **15**

H Firm-owned capital in RBC and the network-firm problem **17**

H.1 Competitive equilibrium 17

H.2 Social Planner's Problem 25

H.3 The adjustment cost 29

A Derivations for the static equilibrium

This appendix derives the static equilibrium conditions reported in Section 2. We assume perfect competition in the simple sector and monopolistic competition for varieties sold to the retailer. All firms are symmetric within each sector.

Marginal costs. The network firm minimizes the cost of the Cobb–Douglas aggregator in the main text. Let λ^N denote unit cost for a network firm and λ^S the unit cost for a simple firm. With competitive input prices and symmetry,

$$\ln \lambda^N = \alpha \ln \lambda^{N,\text{agg}} + (1-\alpha) \ln \lambda^{S,\text{agg}} - \ln A, \tag{1}$$

$$\ln \lambda^S = \ln w - \ln A. \tag{2}$$

Prices. Equating $\lambda^S, \lambda^N, \alpha$ in (2) with $\lambda^{S,\text{agg}}, \lambda^{N,\text{agg}}, \alpha^{\text{agg}}$ since the model features representative firms, we can solve for the sectoral prices as functions of A and w

$$\ln \lambda^{N,\text{agg}} = \ln w - \frac{2-\alpha^{\text{agg}}}{1-\alpha^{\text{agg}}} \ln A, \tag{3}$$

$$\ln \lambda^{S,\text{agg}} = \ln w - \ln A. \tag{4}$$

Retailer's cost minimization. The retailer faces constant-elasticity demand within each block, so the sectoral price indices satisfy

$$P = (p^{N,\text{agg}})^{\zeta} (p^{S,\text{agg}})^{1-\zeta} = \left(\frac{\sigma}{\sigma-1}\right)^{\zeta} (\lambda^{N,\text{agg}})^{\zeta} (\lambda^{S,\text{agg}})^{1-\zeta}. \tag{5}$$

Normalizing $P \equiv 1$ and substituting (1)–(2) yields

$$\ln w = \left(\frac{\zeta}{1-\alpha^{\text{agg}}} + 1\right) \ln A - \zeta \ln \frac{\sigma}{\sigma-1}, \tag{6}$$

which corresponds to the wage equation in the text.

Accounting and profits. Let TC denote total cost of network intermediates used by network firms. A fraction αTC is network goods and $(1-\alpha)TC$ is simple goods. Total cost

of simple goods is wL, so the retailer spends $wL-(1-\alpha)TC$ on simple goods and, by CES expenditure shares, $\frac{\zeta}{1-\zeta}(wL-(1-\alpha)TC)$ on network goods. Converting retailer spending on network goods into costs using the sectoral markup implies

$$\alpha TC+\frac{\zeta}{1-\zeta}\frac{\sigma-1}{\sigma}\big(wL-(1-\alpha)TC\big)=TC, \tag{7}$$

so

$$TC=\frac{\zeta(\sigma-1)}{(1-\alpha)(\sigma-\zeta)}wL. \tag{8}$$

Retailer spending on network goods is then $\frac{\sigma\zeta}{\sigma-\zeta}wL$. Network firms earn revenue from two sources: sales to the retailer at the markup price, and intermediate sales of αTC to other network firms at marginal cost. The latter carry no margin but exactly offset the network-goods fraction of input costs, so network profits equal total revenue minus total input cost,

$$\pi^N=\underbrace{\frac{\sigma\zeta}{\sigma-\zeta}wL}_{\text{retail sales}}+\underbrace{\alpha TC}_{\text{intermediate sales}}-\underbrace{TC}_{\text{input cost}}=\frac{\zeta}{\sigma-\zeta}wL, \tag{9}$$

while perfect competition in the simple sector implies

$$\pi^S=0. \tag{10}$$

Finally, total consumption equals total revenue,

$$C=\frac{\sigma}{\sigma-\zeta}wL=wL+\pi^N, \tag{11}$$

which matches the aggregate consumption expression in the main text.

Calibration mappings. We first derive the intermediate-input share using the effective network intensity $(1-\delta)\alpha_t$ used in period-t production. Substituting $(1-\delta)\alpha_t$ for α in (8),

total intermediate purchases are

$$I_t = TC_t = \frac{\zeta(\sigma-1)}{[1-(1-\delta)\alpha_t]\,(\sigma-\zeta)} w_t L_t. \tag{12}$$

The value of gross output in the network and simple sectors is

$$GO_t^N = \frac{\sigma\zeta}{\sigma-\zeta} w_t L_t + (1-\delta)\alpha_t TC_t, \qquad GO_t^S = w_t L_t. \tag{13}$$

Therefore, $GO_t = GO_t^N + GO_t^S$, and simplification gives

$$\frac{I_t}{GO_t} = \frac{\zeta(\sigma-1)}{\sigma\,[1-(1-\delta)\alpha_t+\zeta]-\zeta}. \tag{14}$$

Next consider upstreamness as defined by Antràs et al. (2012). For sector r, let

$$U_r = 1 + \sum_s \frac{Z_{rs}}{GO_r} U_s,$$

where Z_{rs} is the value of sector-r output used as an intermediate by sector s. In this economy the only intermediate sales are $Z_{NN} = (1-\delta)\alpha_t TC_t$ and $Z_{SN} = [1-(1-\delta)\alpha_t]\,TC_t$. Equations (12) and (13) imply

$$\frac{Z_{NN}}{GO_t^N} = \frac{(1-\delta)\alpha_t(\sigma-1)}{\sigma-(1-\delta)\alpha_t}, \qquad \frac{Z_{SN}}{GO_t^S} = \frac{\zeta(\sigma-1)}{\sigma-\zeta}. \tag{15}$$

The two sectoral upstreamness measures consequently satisfy

$$U_t^N = 1 + \frac{(1-\delta)\alpha_t(\sigma-1)}{\sigma-(1-\delta)\alpha_t} U_t^N = \frac{\sigma-(1-\delta)\alpha_t}{\sigma\,[1-(1-\delta)\alpha_t]}, \tag{16}$$

$$U_t^S = 1 + \frac{\zeta(\sigma-1)}{\sigma-\zeta} U_t^N. \tag{17}$$

Final demand shares are ζ for network goods and $1-\zeta$ for simple goods. Hence final-demand-

weighted upstreamness is

$$\begin{aligned} U_t &= \zeta U_t^N + (1-\zeta) U_t^S \\ &= 1 - \zeta + \frac{\zeta\left[\sigma(2-\zeta)-1\right]}{\sigma-\zeta} \frac{\sigma - (1-\delta)\alpha_t}{\sigma\left[1-(1-\delta)\alpha_t\right]}. \end{aligned} \tag{18}$$

Equivalently, using (14),

$$U_t = 1 - \zeta + \frac{\sigma(2-\zeta)-1}{\sigma-\zeta}\left[\frac{\zeta}{\sigma} + \frac{I_t/GO_t}{1-I_t/GO_t}\right], \tag{19}$$

which is the expression used in the quantitative code. The quantitative targets are the ergodic mean and standard deviation of (14) and the ergodic mean of (18).

B Alternative calibration without targeting output-growth skewness

Alternative calibration set. As a robustness check, we re-run the calibration where skewness of output growth is not targeted. The targeted moments are:

- mean labor supply,
- mean intermediate-input share,
- standard deviation of intermediate-input share,
- standard deviation of HP-filtered log output,
- mean upstreamness.

B.1 Parameter values

Table B.1: Value of parameters and exogenous variables (no-skew calibration)

Symbol	Description	Value	Reference
ρ	Risk aversion	1.000	log utility
β	Discount factor	0.960	annual model frequency
δ	Depreciation rate of network	0.100	annual breaking rate = 10%
σ	Elasticity of substitution	6.000	markup $\approxeq$ 20%
η	Disutility of labor	2.590	joint calibration (Table B.2)
χ	Frisch elasticity of labor	1.000	unit-Frisch benchmark
ζ	Share of network goods	0.821	joint calibration (Table B.2)
μ	Adjustment cost parameter	0.025	joint calibration (Table B.2)
Aggregate fluctuation			
G	TFP value of "good time"	1.006	joint calibration (Table B.2)
B	TFP value of "bad time"	0.997	joint calibration (Table B.2)

Notes: Annual calibration. The parameters $\{\eta, \mu, \zeta, B, G\}$ are jointly calibrated to the moments reported in Table B.2. Remaining parameters are set externally as described in the text.

B.2 Calibration targets and fit

Table B.2: Calibration moments (no-skew calibration)

Moment	Data moment	Model moment
Targeted moments		
Mean labor supply	0.333	0.334
Mean intermediate-input share	0.550	0.550
Standard deviation of intermediate-input share	0.025	0.025
Standard deviation of HP-filtered log output	0.013	0.013
Mean upstreamness	1.780	1.780
Untargeted moments		
Skewness of output growth, 1960–2024	-0.553	-0.167

Notes: Mean labor supply is normalized to one third of available time. The intermediate-input-share targets and moments are from the 1958–2018 NBER–CES Manufacturing Industry Database (Becker, Gray, and Marvakov, 2021). Output volatility is computed from annual U.S. constant-dollar GDP per capita over 1960–2024 from the World Development Indicators, accessed through FRED (series `NYGDPPCAPKDUSA`). The upstreamness target is computed from the BEA input–output tables using the measure of Antràs et al. (2012). Model moments are computed under the calibrated urn–ball adjustment-cost specification using the 10,000 annual observations used for the formal calibration. Output-growth skewness is computed from the corresponding annual output growth series and is not targeted. The model generates about 30% of the data skewness.

B.3 Endogenous-variable means

Table B.3: Ergodic means of endogenous variables (no-skew calibration)

Symbol	Description	Ergodic mean
α	Network intensity	0.487
w	Wage	0.864
N	Labor	0.334
C	Consumption	0.334
Y	Output	0.334

Notes: Ergodic means computed from the same 10,000 annual observations used for the model moments in Table B.2.

By construction, this calibration improves the fit to the intermediate-input volatility relative to the main-text baseline calibration that targets output-growth skewness. In the baseline calibration, the model fit misses the targeted standard deviation of the aggregate intermediate-input share: 0.0771 instead of 0.025. In this alternative calibration, the model generates about 30.2% of the left-skewness observed in U.S. output growth.

C Simulation Exercises

C.1 Ergodic moments

Table C.4 reports summary statistics of the endogenous variables along the simulated ergodic path.

Table C.4: Ergodic means of endogenous variables

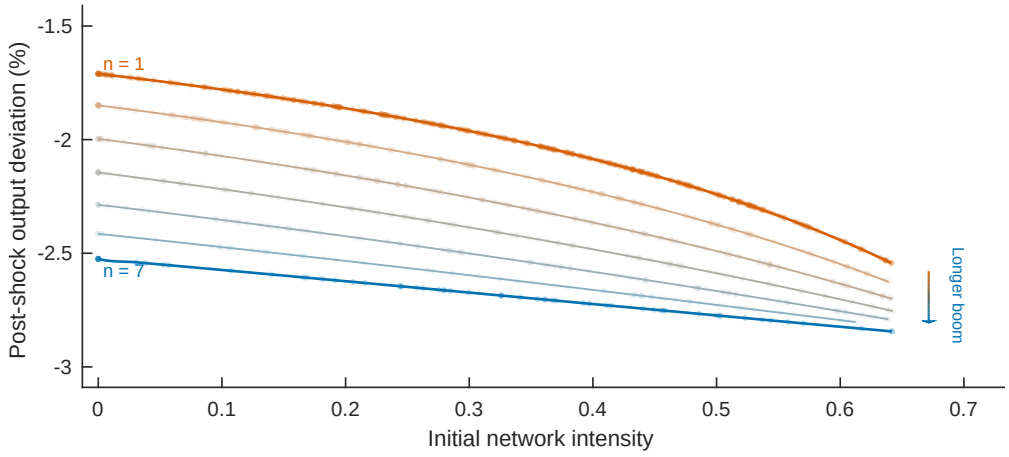
Symbol	Description	Ergodic mean
α	Network intensity	0.434
w	Wage	0.856
N	Labor	0.334
C	Consumption	0.333
Y	Output	0.333

Notes: Ergodic means use the same 9500 annual observations as the calibration-moments table in the main text.

C.2 Joint size and duration dependence

Figure C.1 reports the joint dependence of the shock response on the inherited network state and the length of the preceding boom.

Figure C.1: Joint size and duration dependence in output shock responses

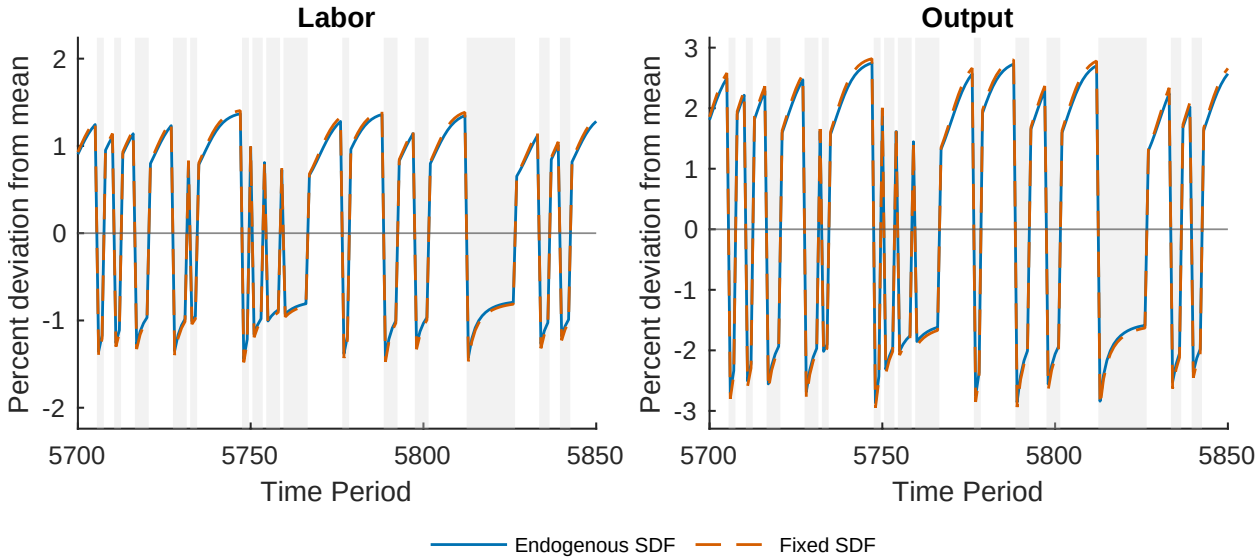


Notes: Each curve plots output immediately after a bad shock following n consecutive good states, $Y(g^n(\alpha, G), B)$, against the initial network state α at the start of the good spell. Output is expressed as a percentage deviation from its simulated mean. The orange and blue curves correspond to $n = 1$ and $n = 7$; the muted intermediate curves correspond to $n = 2, \ldots, 6$, ordered from top to bottom. Within each n, the curve slopes down (size dependence), while the downward shift as n rises shows duration dependence.

C.3 Fixed SDF

Figure C.2 compares the benchmark equilibrium with a counterfactual in which the stochastic discount factor is fixed at $M = \beta$, while all other equilibrium conditions are re-solved. The resulting paths are nearly identical. Fixing the stochastic discount factor raises the standard deviation of HP-filtered log output only from 1.300 percent to 1.344 percent and leaves the timing and asymmetry of fluctuations essentially unchanged. Thus, the endogenous stochastic discount factor modestly dampens aggregate volatility but is not central to the plucking mechanism.

Figure C.2: Benchmark and fixed-SDF counterfactual



Notes: Both economies are simulated under the same productivity sequence. The lines report percentage deviations of labor and output from their respective simulated means over indices 5,700–5,850. Dark-blue solid lines represent the benchmark equilibrium with an endogenous stochastic discount factor; orange dashed lines represent the counterfactual with $M = \beta$. Gray shaded areas indicate bad productivity states ($A = B$).

D A plucking-cycle exercise in simulated output

This appendix implements an analogous plucking-cycle exercise in the simulated model, using the threshold turning-point algorithm in Dupraz, Nakamura, and Steinsson (2025). The exercise asks whether the model-generated output series displays the one-sided predictability pattern emphasized in the plucking literature: contraction amplitudes predict subsequent recoveries, while recovery amplitudes have little predictive power for subsequent contractions.

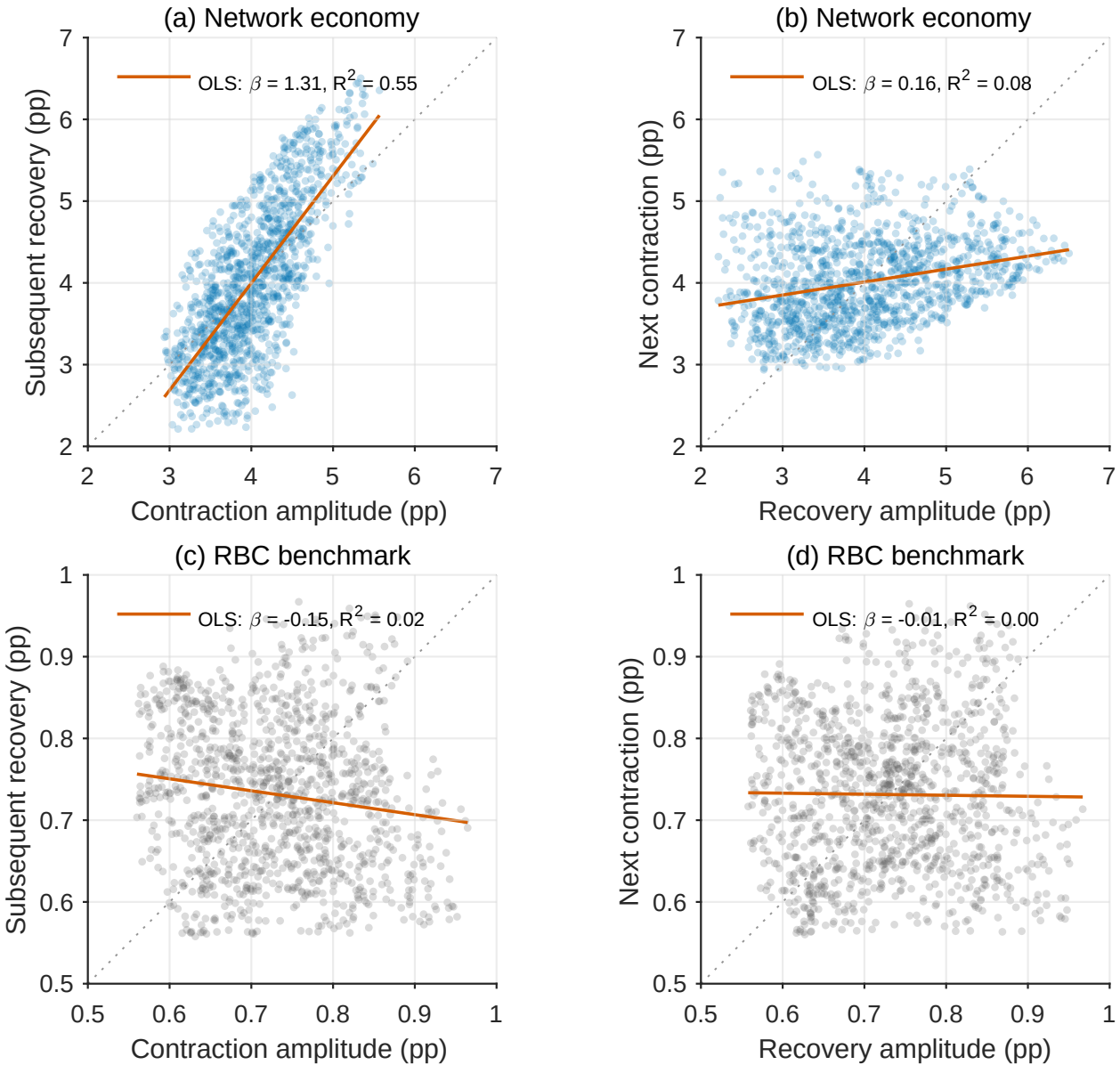
The empirical implementation in Dupraz, Nakamura, and Steinsson (2025) identifies peaks and troughs in unemployment in levels. Since the algorithm is designed for unemployment, where higher values indicate greater slack, we transform simulated output into a comparable shortfall measure. Consistent with Friedman's ceiling interpretation, we measure output relative to a smooth moving ceiling and define

$$u_t = -\widehat{y}_t, \qquad \widehat{y}_t = \log Y_t - \mathrm{HP}(\log Y_t),$$

where the HP smoothing parameter is $\lambda = 6.25$ for annual data. We then apply the threshold turning-point logic to u_t, with the cycle-size threshold set to 1.5 standard deviations of u_t.

For each identified cycle, we compute the initial contraction amplitude and the subsequent recovery amplitude from the output-gap series $\widehat{y}_t$.

Figure D.3: Plucking-cycle exercise for simulated output slack



Notes: The figure applies the turning-point algorithm in Dupraz, Nakamura, and Steinsson (2025) to simulated model output slack, $u_t = -\widehat{y}_t$, where $\widehat{y}_t$ is the HP-filtered log output gap. The cycle-size threshold is 1.5 standard deviations of u_t. Amplitudes are expressed as percent changes in output relative to the smooth ceiling. The left panel regresses subsequent recovery amplitudes on initial contraction amplitudes. The right panel regresses subsequent contraction amplitudes on prior recovery amplitudes.

Figure D.3 reports the two plucking regressions. The left panel regresses the subsequent recovery amplitude on the initial contraction amplitude. The right panel regresses the sub-

sequent contraction amplitude on the prior recovery amplitude. In the simulated network economy, contractions strongly predict subsequent recoveries, with an R^2 of 0.49. In contrast, prior recoveries explain little of the next contraction, with an R^2 of 0.06. This one-sided predictability is consistent with the model's plucking interpretation: downturns represent stochastic drops below a moving ceiling, while recoveries reflect within-regime convergence back toward that ceiling. Hence recovery is tied to the size of the inherited shortfall, whereas the next downturn is governed by a new aggregate-state realization rather than mechanically by the size of the previous recovery.

We can also implement the exercise without applying any statistical filter. Let the regime-specific attractor be output evaluated at the conditional saddle $\alpha_j^* = g_j(\alpha_j^*)$ for productivity regime A_j, and decompose log output as

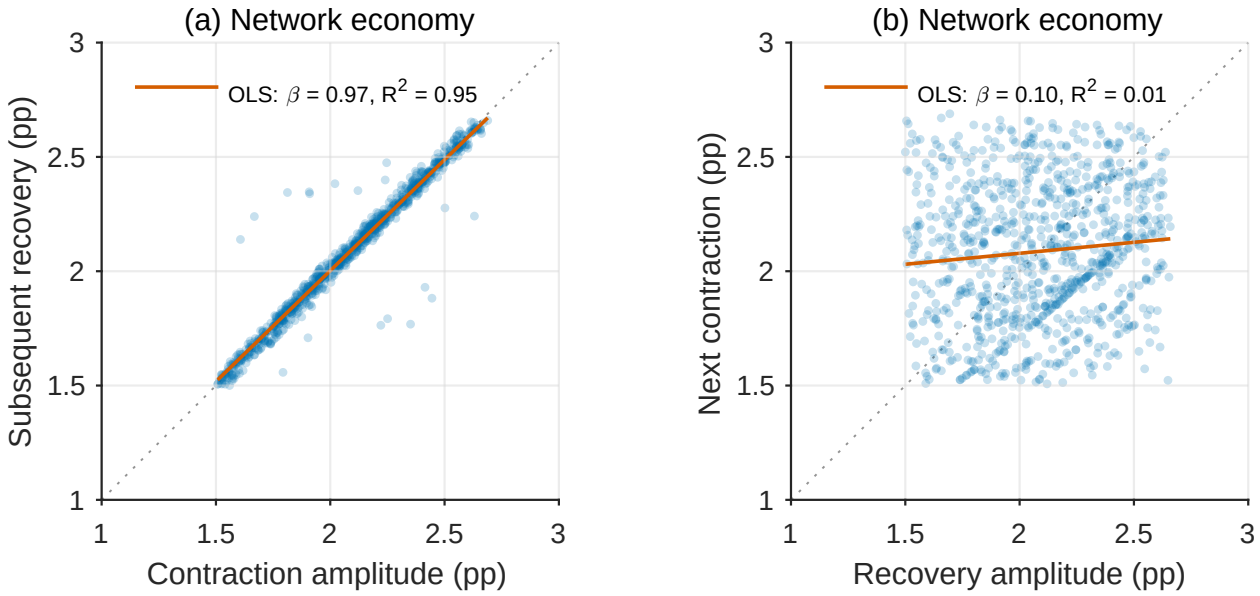
$$\log Y_t = \log Y^*(A_t) + c_t^{\text{pluck}}, \qquad c_t^{\text{pluck}} \equiv \log Y_t - \log Y^*(A_t). \tag{20}$$

This is the conditional-saddle construction developed in the main text, and c_t^{pluck} is therefore a model-implied, unfiltered deviation from the relevant regime-specific attractor. Figure D.4 applies the same turning-point procedure directly to this structural component. It produces the same predictability asymmetry as the HP-based measure: contractions tightly predict subsequent recoveries, whereas recoveries contain essentially no information about the next contraction. In a two-state calibration, the exercise can generate near-mechanical contraction–recovery correlation from the fixed two-attractor geometry. Using three states avoids that mechanical channel and reports the structural asymmetry in the richer state space.

E Additional empirical results

In this section, we present additional empirical results that further support the model's mechanism and its implications for sectoral employment dynamics during recessions. Table E.5 reports the regressions of Table 5 with material share and boom duration entered separately and jointly, for employment, payroll and gross output, in the data (Panel A) and in the

Figure D.4: Predictability asymmetry in the model-implied plucking component



Notes: The plotted series is $c_t^{\text{pluck}} = \log Y_t - \log Y^*(A_t)$, where $Y^*(A_t)$ is output at the regime-specific conditional saddle. No statistical filter is applied. In a two-state economy this construction can deliver near-mechanical contraction–recovery comovement, so we use a three-state Rouwenhorst process here, and turning points are dated with a 1.5 percent directional-change threshold. The left panel plots each recovery against the preceding contraction; the right panel plots the next contraction against the preceding recovery. The estimated slopes are 0.969 and 0.096, respectively, with R^2 values of 0.954 and 0.009.

model (Panel B). Columns (7)–(9) reproduce Table 5. The material-share coefficient is positive and significant for every outcome, alone and conditional on boom duration. Table E.6 repeats the data regressions with the non-energy material share, (matcost − energy)/vship; the estimates are similar.

F Canonical RBC model for benchmark

This appendix describes the canonical RBC model used to construct the benchmark conditional saddle paths in Figure 5(a). Throughout this appendix, α denotes the capital share and δ the capital depreciation rate, following standard RBC convention; these symbols are distinct from the network intensity and the network depreciation rate of the main text. A

Table E.5: Material share and peak-to-trough drop: separate and joint regressors

Drop^Y, Y=	(1) emp	(2) pay	(3) vship	(4) emp	(5) pay	(6) vship	(7) emp	(8) pay	(9) vship
Panel A: Data									
Material Cost Share	0.132**	0.143**	0.167**				0.120**	0.123**	0.141**
	(0.040)	(0.044)	(0.051)				(0.043)	(0.045)	(0.054)
Booming Duration				0.014***	0.011***	0.014***	0.014***	0.011***	0.014***
				(0.002)	(0.002)	(0.003)	(0.002)	(0.002)	(0.003)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Peak year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.349	0.352	0.311	0.363	0.365	0.325	0.365	0.367	0.326
N	5882	5882	5882	5518	5518	5518	5518	5518	5518
Panel B: Model									
Material Cost Share	0.0994***	0.1987***	0.5030***				0.0962***	0.1924***	0.2727***
	(0.0022)	(0.0044)	(0.0241)				(0.0025)	(0.0049)	(0.0226)
Booming Duration				0.0025***	0.0049***	0.0409***	0.0005***	0.0010***	0.0353***
				(0.0002)	(0.0004)	(0.0015)	(0.0001)	(0.0003)	(0.0015)
R-squared	0.394	0.394	0.122	0.077	0.077	0.256	0.397	0.397	0.287
N	2318	2318	2318	2317	2317	2317	2317	2317	2317

Notes: The dependent variable is the log peak-to-trough drop in employment, payroll, or the value of shipments (vship). Peaks and troughs are adjacent local extrema of the HP-filtered level of employment, and material share is measured at the peak. Panel A includes industry and peak-year fixed effects and peak capital stock; standard errors are clustered by industry. Panel B applies the same local-extrema construction to the calibrated model's ergodic sample and reports heteroskedasticity-robust standard errors. Columns (7)–(9) are Table 5 in the main text. Standard errors are in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

representative household solves

$$v(a; K, A) = \max_{c,a'} \quad \log(c) + \beta \mathbb{E}\left[v(a'; K', A') \mid A\right] \tag{21}$$

$$\text{s.t.} \quad c + a' = a(1 + r(K, A)) + w(K, A) \tag{22}$$

$$a' \geq 0, \tag{23}$$

where v is the value function, a is wealth at the beginning of the period, K is aggregate capital, and A is aggregate TFP. The aggregate state evolves according to

$$K' = \Gamma_{\text{endo}}(K, A), \qquad A' \sim \Gamma_{\text{exo}}(\cdot \mid A), \tag{24}$$

Table E.6: Non-energy material share and peak-to-trough drop

Drop^Y, Y=	(1) emp	(2) pay	(3) vship	(4) emp	(5) pay	(6) vship	(7) emp	(8) pay	(9) vship
Material Cost Share	0.123**	0.137**	0.179***				0.107*	0.115*	0.153**
	(0.041)	(0.045)	(0.051)				(0.044)	(0.046)	(0.054)
Booming Duration				0.014***	0.011***	0.014***	0.014***	0.011***	0.013***
				(0.002)	(0.002)	(0.003)	(0.002)	(0.002)	(0.003)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Peak year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.348	0.352	0.311	0.363	0.365	0.325	0.365	0.367	0.327
N	5882	5882	5882	5518	5518	5518	5518	5518	5518

Notes: The dependent variable is the log peak-to-trough drop in employment, payroll, or the value of shipments (vship). Peaks and troughs are adjacent local extrema of the HP-filtered level of employment, and the non-energy material share, (matcost − energy)/vship, is measured at the peak. Industry and peak-year fixed effects and peak capital stock are included; standard errors are clustered by industry. Standard errors are in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

where Γ_{endo} is the equilibrium law of motion for capital. TFP follows a two-state Markov chain between $A \in \{B, G\}$ with $G > B$ and transition matrix

$$\Gamma_{\text{exo}} = \begin{bmatrix} \pi_{BB} & \pi_{BG} \\ \pi_{GB} & \pi_{GG} \end{bmatrix}. \tag{25}$$

Production is Cobb-Douglas with constant returns to scale. Competitive factor prices are

$$r(K, A) = \alpha A K^{\alpha-1} - \delta, \tag{26}$$

$$w(K, A) = (1 - \alpha) A K^{\alpha}, \tag{27}$$

where δ is the depreciation rate. In equilibrium, the representative household's asset holdings equal the aggregate capital stock: $a = K$.

G CES extensions for unit costs

This appendix checks how the static sign result

$$\frac{\partial \ln \lambda^N}{\partial \ln A} < 0, \qquad \frac{\partial \ln \lambda^S}{\partial \ln A} > 0$$

extends beyond Cobb–Douglas aggregation. Throughout, hold aggregate network intensity fixed at $\alpha \in (0,1)$ and normalize the final-good price index to one. The Cobb–Douglas cases $\varepsilon_F = 1$ and $\varepsilon_N = 1$ are understood as limits.

CES final sector. First keep the benchmark network-sector unit-cost equation:

$$\ln \lambda^N = \alpha \ln \lambda^N + (1-\alpha) \ln \lambda^S - \ln A.$$

Then

$$\frac{\lambda^N}{\lambda^S} = A^{-1/(1-\alpha)}.$$

Now let the final sector be CES. The network-sector markup enters through the sector price $(\sigma/(\sigma-1))\lambda^N$:

$$1 = \left[\zeta \left(\frac{\sigma}{\sigma - 1} \lambda^N \right)^{1-\varepsilon_F} + (1-\zeta)(\lambda^S)^{1-\varepsilon_F} \right], \qquad \zeta \in (0,1). \tag{28}$$

Substituting the relative-cost equation into (28) gives

$$1 = \left[\zeta \left(\frac{\sigma}{\sigma - 1} \right)^{1-\varepsilon_F} A^{-(1-\varepsilon_F)/(1-\alpha)} + 1 - \zeta \right] (\lambda^S)^{1-\varepsilon_F}.$$

Therefore,

$$\begin{aligned} \frac{\partial \ln \lambda^S}{\partial \ln A} &= \frac{1}{1-\alpha} \frac{\zeta \left(\frac{\sigma}{\sigma-1}\right)^{1-\varepsilon_F} A^{-(1-\varepsilon_F)/(1-\alpha)}}{\zeta \left(\frac{\sigma}{\sigma-1}\right)^{1-\varepsilon_F} A^{-(1-\varepsilon_F)/(1-\alpha)} + 1 - \zeta}, \\ \frac{\partial \ln \lambda^N}{\partial \ln A} &= -\frac{1}{1-\alpha} \frac{1-\zeta}{\zeta \left(\frac{\sigma}{\sigma-1}\right)^{1-\varepsilon_F} A^{-(1-\varepsilon_F)/(1-\alpha)} + 1 - \zeta}. \end{aligned} \tag{29}$$

Evaluated at $A = 1$,

$$\left.\frac{\partial \ln \lambda^S}{\partial \ln A}\right|_{A=1} = \frac{1}{1-\alpha}\frac{\zeta\left(\frac{\sigma}{\sigma-1}\right)^{1-\varepsilon_F}}{\zeta\left(\frac{\sigma}{\sigma-1}\right)^{1-\varepsilon_F}+1-\zeta} > 0,$$
$$\left.\frac{\partial \ln \lambda^N}{\partial \ln A}\right|_{A=1} = -\frac{1}{1-\alpha}\frac{1-\zeta}{\zeta\left(\frac{\sigma}{\sigma-1}\right)^{1-\varepsilon_F}+1-\zeta} < 0. \tag{30}$$

Thus the final-sector curvature changes the expenditure-share term away from $A = 1$, but not the signs. In this case the sign result is global for $A > 0$.

CES network sector. Now keep the final sector Cobb–Douglas:

$$1 = \left(\frac{\sigma}{\sigma-1}\lambda^N\right)^{\zeta}(\lambda^S)^{1-\zeta}.$$

Let the network-sector unit cost be CES, with A entering as a level Hicks-neutral productivity shifter:

$$\lambda^N = \frac{1}{A}\left[\alpha(\lambda^N)^{1-\varepsilon_N} + (1-\alpha)(\lambda^S)^{1-\varepsilon_N}\right]^{1/(1-\varepsilon_N)}.$$

Raising both sides to $1-\varepsilon_N$ and rearranging gives

$$\left(\frac{\lambda^N}{\lambda^S}\right)^{1-\varepsilon_N} = \frac{1-\alpha}{A^{1-\varepsilon_N}-\alpha}.$$

This expression also makes clear the admissible region for A under the CES network-sector unit cost. Since $\lambda^N/\lambda^S > 0$, the denominator on the right-hand side must be positive:

$$A^{1-\varepsilon_N} > \alpha.$$

This restriction is automatically satisfied at $A = 1$. Away from $A = 1$, it can bind: when $\varepsilon_N < 1$, it rules out sufficiently low A; when $\varepsilon_N > 1$, it rules out sufficiently high A. We therefore interpret this exercise locally around $A = 1$, or on any support of A contained in the admissible region. For A satisfying this condition,

$$\frac{\partial \ln(\lambda^N/\lambda^S)}{\partial \ln A} = -\frac{A^{1-\varepsilon_N}}{A^{1-\varepsilon_N}-\alpha}.$$

The Cobb–Douglas final-sector normalization implies

$$d\ln\lambda^S = -\zeta\, d\ln(\lambda^N/\lambda^S), \qquad d\ln\lambda^N = (1-\zeta)\, d\ln(\lambda^N/\lambda^S).$$

Therefore

$$\begin{aligned}\frac{\partial \ln\lambda^S}{\partial \ln A} &= \zeta\frac{A^{1-\varepsilon_N}}{A^{1-\varepsilon_N}-\alpha},\\ \frac{\partial \ln\lambda^N}{\partial \ln A} &= -(1-\zeta)\frac{A^{1-\varepsilon_N}}{A^{1-\varepsilon_N}-\alpha}.\end{aligned} \tag{31}$$

At $A = 1$,

$$\left.\frac{\partial \ln\lambda^S}{\partial \ln A}\right|_{A=1} = \frac{\zeta}{1-\alpha} > 0, \qquad \left.\frac{\partial \ln\lambda^N}{\partial \ln A}\right|_{A=1} = -\frac{1-\zeta}{1-\alpha} < 0.$$

Thus, with a standard Hicks-neutral productivity shifter in the CES unit-cost function, the sign result is unchanged for both complements and substitutes locally around $A = 1$.

H Firm-owned capital in RBC and the network-firm problem

This appendix compares the value-maximization problem of a capital-owning firm in standard RBC with the corresponding problem of a network firm. We first write each problem in primitive form and list its first-order conditions before comparing private and social derivatives.

H.1 Competitive equilibrium

Final sector's problem. The final sector is static. It purchases X_j^N and X_j^S from the network and simple sectors, respectively, and sells Y to the household at a price normalized to one. It takes all final-market input prices $\{p_j^{N,\text{final}}, p_j^{S,\text{final}}\}_{j\in[0,1]}$ as given and chooses

quantities. Its problem is

$$\Pi^F(\mathcal{X}) = \max_{Y,\{X_j^N\},\{X_j^S\}} \left\{ Y - \int_0^1 p_j^{N,\text{final}} X_j^N \, \mathrm{d}j - \int_0^1 p_j^{S,\text{final}} X_j^S \, \mathrm{d}j \right\}$$
$$\text{subject to} \qquad Y = \frac{1}{\zeta^\zeta (1-\zeta)^{1-\zeta}} \left[\left(\int_0^1 (X_j^N)^{\frac{\sigma-1}{\sigma}} \, \mathrm{d}j \right)^{\frac{\sigma}{\sigma-1}} \right]^\zeta \left[\exp\left(\int_0^1 \ln(X_j^S) \, \mathrm{d}j \right) \right]^{1-\zeta}. \tag{32}$$

Let λ^F be the multiplier on the production constraint. The first-order conditions are

$$[X_j^N]: \qquad p_j^{N,\text{final}} = \lambda^F \zeta Y \frac{(X_j^N)^{-1/\sigma}}{\int_0^1 (X_{j'}^N)^{\frac{\sigma-1}{\sigma}} \, \mathrm{d}j'}, \tag{33}$$

$$[X_j^S]: \qquad p_j^{S,\text{final}} = \lambda^F (1-\zeta) \frac{Y}{X_j^S}, \tag{34}$$

$$[Y]: \qquad 1 = \lambda^F. \tag{35}$$

The corresponding sectoral price indices are

$$p^{N,\text{agg}} = \left(\int_0^1 (p_j^{N,\text{final}})^{1-\sigma} \, \mathrm{d}j \right)^{\frac{1}{1-\sigma}}, \qquad p^{S,\text{agg}} = \exp\left(\int_0^1 \ln p_j^{S,\text{final}} \, \mathrm{d}j \right). \tag{36}$$

These expressions follow from the final-sector first-order conditions. Raising (33) to the power $1-\sigma$ and integrating over j, and taking logs of (34) and integrating over j, give

$$p^{N,\text{agg}} = \lambda^F \zeta Y \left(\int_0^1 (X_j^N)^{\frac{\sigma-1}{\sigma}} \, \mathrm{d}j \right)^{-\frac{\sigma}{\sigma-1}}, \quad p^{S,\text{agg}} = \lambda^F (1-\zeta) Y \exp\left(-\int_0^1 \ln X_j^S \, \mathrm{d}j \right). \tag{37}$$

Taking the weighted product of these two equations and using the production constraint in (32) yields the unit price of final output:

$$P = \left(p^{N,\text{agg}}\right)^\zeta \left(p^{S,\text{agg}}\right)^{1-\zeta} = \lambda^F = 1, \tag{38}$$

where the last equality is the final-good price normalization, equivalently the Y first-order condition (35). No condition on network firms' pricing has been imposed at this stage.

Using $\lambda^F = 1$, the network-sector first-order condition yields the demand curve

$$X_j^N = \frac{\zeta Y}{p^{N,\text{agg}}} \left(\frac{p_j^{N,\text{final}}}{p^{N,\text{agg}}} \right)^{-\sigma}. \tag{39}$$

Thus the demand curve for each network firm is given by

$$\frac{\partial \ln X_j^N}{\partial \ln p_j^{N,\text{final}}} = -\sigma. \tag{40}$$

Simple firms' problem. The simple firms' problem is static. Firm j hires labor, sells X_j^S to the final sector, and sells

$$\int_0^1 x_{ji}^S \mathbf{1}\{i \text{ buys from } j\} \, \mathrm{d}i$$

to network firms. Here the indicator means that simple variety j belongs to network buyer i's simple-input bundle; quantities are chosen only for such active orders. The firm is a price taker in both sales markets and in the labor market: it takes $p_j^{S,\text{final}}$, $p_j^{S,\text{int}}$, and w as given. The firm's problem is

$$\begin{aligned} \Pi_j^S(\mathcal{X}) = \max_{\ell_j, X_j^S, \{x_{ji}^S\}} & \left\{ p_j^{S,\text{final}} X_j^S + p_j^{S,\text{int}} \int_0^1 x_{ji}^S \mathbf{1}\{i \text{ buys from } j\} \, \mathrm{d}i - w\ell_j \right\} \\ \text{subject to} \qquad & X_j^S + \int_0^1 x_{ji}^S \mathbf{1}\{i \text{ buys from } j\} \, \mathrm{d}i \le A\ell_j = y_j^S. \end{aligned} \tag{41}$$

Let λ_j^S be the multiplier on the production constraint. The first-order conditions are

$$\begin{aligned} [X_j^S]: \qquad & p_j^{S,\text{final}} = \lambda_j^S, \\ [x_{ji}^S,\ i \text{ buys from } j]: \qquad & p_j^{S,\text{int}} = \lambda_j^S, \\ [\ell_j]: \qquad & w = \lambda_j^S A. \end{aligned} \tag{42}$$

Network firms' problem. Firm j sells X_j^N to the retailer and

$$\int_0^1 x_{ji}^N \mathbf{1}\{i \text{ buys from } j\} \, \mathrm{d}i$$

to other network firms, as in the network-goods market-clearing condition in the main text. It purchases x_{ij}^N and x_{kj}^S as inputs. The retailer quantity $X_j^N(p_j^{N,\text{final}}, \mathcal{X})$ is the CES demand induced by the final-market price $p_j^{N,\text{final}}$ chosen by firm j, given the aggregate state $\mathcal{X} = [\alpha^{\text{agg}}, A]$. The household stochastic discount factor is

$$M(\mathcal{X}, \mathcal{X}') = \beta \frac{u_c(c', n')}{u_c(c, n)}.$$

The firm is a price taker in every intermediate-good market. In particular, it takes its intermediate sales price $p_j^{N,\text{int}}$, the input prices $\{p_i^{N,\text{int}}\}$ and $\{p_k^{S,\text{int}}\}$, and the perceived law of motion for $\mathcal{X}'$ as given. The only price it chooses is $p_j^{N,\text{final}}$. Its value problem is

$$\begin{aligned} J(\alpha; \mathcal{X}) = \max_{\substack{\alpha' \in [0,1),\, p_j^{N,\text{final}}, \\ \{x_{ij}^N\},\, \{x_{kj}^S\}}} \Bigg\{ & \\ & p_j^{N,\text{final}} X_j^N(p_j^{N,\text{final}}, \mathcal{X}) + p_j^{N,\text{int}} \int_0^1 x_{ji}^N \mathbf{1}\{i \text{ buys from } j\} \, \mathrm{d}i \\ & - \int_0^{(1-\delta)\alpha} p_i^{N,\text{int}} x_{ij}^N \, \mathrm{d}i - \int_{(1-\delta)\alpha}^1 p_k^{S,\text{int}} x_{kj}^S \, \mathrm{d}k \\ & - \Phi(\alpha, \alpha') + \mathbb{E}[M(\mathcal{X}, \mathcal{X}') J(\alpha'; \mathcal{X}')] \Bigg\} \end{aligned} \tag{43}$$

$$\begin{aligned} \text{subject to} \quad & X_j^N(p_j^{N,\text{final}}, \mathcal{X}) + \int_0^1 x_{ji}^N \mathbf{1}\{i \text{ buys from } j\} \, \mathrm{d}i \\ & \leq A \exp\left(\int_0^{(1-\delta)\alpha} \ln x_{ij}^N \, \mathrm{d}i + \int_{(1-\delta)\alpha}^1 \ln x_{kj}^S \, \mathrm{d}k \right) = y_j^N, \\ & (\alpha^{\text{agg}})' = \Gamma_{\text{endo}}(\mathcal{X}). \end{aligned}$$

The multiplier on the production constraint is the firm's marginal cost, so we use the

existing notation λ^N for it.

$$[p_j^{N,\text{final}}]: \qquad 0 = X_j^N(p_j^{N,\text{final}}, \mathcal{X}) + \left[p_j^{N,\text{final}} - \lambda^N\right] \frac{\partial X_j^N(p_j^{N,\text{final}}, \mathcal{X})}{\partial p_j^{N,\text{final}}},$$

$$[\text{using (40)}]: \qquad p_j^{N,\text{final}} = \frac{\sigma}{\sigma - 1}\lambda^N, \tag{44}$$

$$\left[x_{ij}^N, i \in [0, (1-\delta)\alpha]\right] \qquad 0 = -p_i^{N,\text{int}} + \lambda^N \frac{y_j^N}{x_{ij}^N}, \tag{45}$$

$$\left[x_{kj}^S, k \in [(1-\delta)\alpha, 1]\right] \qquad 0 = -p_k^{S,\text{int}} + \lambda^N \frac{y_j^N}{x_{kj}^S}, \tag{46}$$

$$[\alpha'] \qquad 0 = -\Phi_2(\alpha, \alpha') + \mathbb{E}\left[M(\mathcal{X}, \mathcal{X}') \frac{\partial J(\alpha'; \mathcal{X}')}{\partial \alpha'}\right], \tag{47}$$

$$[\lambda^N] \qquad 0 = A \exp\left(\int_0^{(1-\delta)\alpha} \ln x_{ij}^N \, \mathrm{d}i + \int_{(1-\delta)\alpha}^1 \ln x_{kj}^S \, \mathrm{d}k\right) - X_j^N(p_j^{N,\text{final}}, \mathcal{X}) - \int_0^1 x_{ji}^N \mathbf{1}\{i \text{ buys from } j\} \, \mathrm{d}i. \tag{48}$$

Envelope condition.

$$\frac{\partial J(\alpha; \mathcal{X})}{\partial \alpha} = (1-\delta)\Big\{ - p_i^{N,\text{int}} x_{ij}^N\Big|_{i=(1-\delta)\alpha} + p_k^{S,\text{int}} x_{kj}^S\Big|_{k=(1-\delta)\alpha} + \lambda^N y_j^N \left[\ln x_{ij}^N\Big|_{i=(1-\delta)\alpha} - \ln x_{kj}^S\Big|_{k=(1-\delta)\alpha}\right]\Big\} - \Phi_1(\alpha, \alpha'). \tag{49}$$

The three terms in braces come, respectively, from the moving upper cost limit, the moving lower cost limit, and the production constraint.

Evaluate the input conditions (45) and (46) at the moving boundary:

$$\begin{aligned} p_i^{N,\text{int}} x_{ij}^N\Big|_{i=(1-\delta)\alpha} &= \lambda^N y_j^N, \\ p_k^{S,\text{int}} x_{kj}^S\Big|_{k=(1-\delta)\alpha} &= \lambda^N y_j^N. \end{aligned} \tag{50}$$

Consequently, the first two terms in (49) cancel. The same two conditions imply

$$\frac{x_{ij}^N\big|_{i=(1-\delta)\alpha}}{x_{kj}^S\big|_{k=(1-\delta)\alpha}} = \frac{p_k^{S,\text{int}}\Big|_{k=(1-\delta)\alpha}}{p_i^{N,\text{int}}\Big|_{i=(1-\delta)\alpha}}. \tag{51}$$

Substituting these results into (49) gives

$$\frac{\partial J(\alpha;\mathcal{X})}{\partial \alpha} = (1-\delta)\lambda^N y_j^N \ln\left(\frac{p_k^{S,\text{int}}\Big|_{k=(1-\delta)\alpha}}{p_i^{N,\text{int}}\Big|_{i=(1-\delta)\alpha}}\right) - \Phi_1(\alpha,\alpha'). \tag{52}$$

Symmetric equilibrium. We can now evaluate the private derivative at the symmetric equilibrium. Impose $\alpha = \alpha^{\text{agg}}$. The symmetric sourcing pattern implies that every simple firm supplies the same measure of network buyers, $\int_0^1 \mathbf{1}\{i \text{ buys from } j\}\,\mathrm{d}i = 1-(1-\delta)\alpha$. Because network goods sold as intermediate inputs are traded competitively, their price equals the network firm's marginal cost at symmetry: $p_j^{N,\text{int}} = \lambda^N$. The final-market pricing condition (44), the network price-index definition in (36), and symmetry across network firms then imply

$$p^{N,\text{agg}} = p_j^{N,\text{final}} = \frac{\sigma}{\sigma-1} p_j^{N,\text{int}}. \tag{53}$$

The two sales conditions in (42) give $p_j^{S,\text{final}} = p_j^{S,\text{int}} = \lambda_j^S$. Symmetry across simple firms and the simple-sector price-index definition then imply $p^{S,\text{agg}} = p_j^{S,\text{final}} = p_j^{S,\text{int}} = \lambda_j^S$. Only at this point can the final-price normalization (38) be written in terms of the two marginal costs:

$$1 = \left(\frac{\sigma}{\sigma-1} p_j^{N,\text{int}}\right)^{\zeta} \left(p_j^{S,\text{int}}\right)^{1-\zeta}. \tag{54}$$

We next derive their relative price from the network technology. Using $\lambda^N = p_j^{N,\text{int}}$, the two input conditions (45) and (46) give, respectively,

$$x_{ij}^N = \frac{\lambda^N}{p_j^{N,\text{int}}} y_j^N = y_j^N, \qquad x_{kj}^S = \frac{\lambda^N}{p_j^{S,\text{int}}} y_j^N = \frac{p_j^{N,\text{int}}}{p_j^{S,\text{int}}} y_j^N. \tag{55}$$

Substituting both expressions into the binding production constraint (48) and dividing by y_j^N yields

$$1 = A\left(\frac{p_j^{N,\text{int}}}{p_j^{S,\text{int}}}\right)^{1-(1-\delta)\alpha},$$
$$\ln\left(\frac{p_j^{S,\text{int}}}{p_j^{N,\text{int}}}\right) = \frac{\ln A}{1-(1-\delta)\alpha}. \tag{56}$$

Combining this ratio with the final-price normalization (54) pins down the two price levels:

$$p_j^{N,\text{int}} = \left(\frac{\sigma}{\sigma-1}\right)^{-\zeta} A^{-\frac{1-\zeta}{1-(1-\delta)\alpha}}, \qquad p_j^{S,\text{int}} = \left(\frac{\sigma}{\sigma-1}\right)^{-\zeta} A^{\frac{\zeta}{1-(1-\delta)\alpha}}. \tag{57}$$

We can now express the envelope directly in terms of final output. At symmetry, the final-sector network-input condition (33) and $\lambda^F = 1$ imply

$$p_j^{N,\text{final}} X_j^N = \zeta Y.$$

Using the retail pricing relation (53),

$$\lambda^N X_j^N = \frac{\sigma-1}{\sigma}\zeta Y.$$

Network-goods market clearing gives $X_j^N = [1-(1-\delta)\alpha]y_j^N$. Therefore,

$$\lambda^N y_j^N = \frac{\sigma-1}{\sigma}\frac{\zeta Y}{1-(1-\delta)\alpha}. \tag{58}$$

Substituting (58) and (56) into (52) yields the symmetric-equilibrium envelope condition:

$$\frac{\partial J(\alpha;\mathcal{X})}{\partial \alpha} = \frac{\sigma-1}{\sigma}\zeta Y \ln A \frac{1-\delta}{[1-(1-\delta)\alpha]^2} - \Phi_1(\alpha,\alpha'). \tag{59}$$

Substitution into the α' condition. Evaluate (59) next period and substitute it into (47). The symmetric-equilibrium CE Euler equation is

$$\Phi_2(\alpha, \alpha') = \mathbb{E}\left[M(\mathcal{X}, \mathcal{X}')\left\{\frac{\sigma - 1}{\sigma}\zeta Y' \ln A' \frac{1-\delta}{\left[1-(1-\delta)\alpha'\right]^2} - \Phi_1(\alpha', \alpha'')\right\}\right]. \tag{60}$$

H.2 Social Planner's Problem

The social planner chooses all quantities directly, including final demand X_j^N for each network variety. Prices therefore do not enter the planner's problem. The planner is constrained only by the production technologies, resource feasibility, and the law of motion for network intensity.

Let $a_j \equiv (1-\delta)\alpha_j$. Before imposing symmetry, the state contains the entire profile $\{\alpha_j\}_{j\in[0,1]}$. The recursive objective is

$$V^{SP}(\{\alpha_j\}, A) = \max_{\substack{c,n,Y,\{\alpha_j',\ell_j,X_j^N,X_j^S\}_{j\in[0,1]},\\ \{x_{ij}^N,x_{kj}^S\}_{i,k,j\in[0,1]}}} \left\{u(c,n) + \beta\mathbb{E}\left[V^{SP}\left(\{\alpha_j'\}, A'\right)\middle|A\right]\right\} \tag{61}$$

$$\text{subject to} \qquad c = Y - \int_0^1 \Phi(\alpha_j, \alpha_j')\,\mathrm{d}j, \tag{62}$$

$$Y = \frac{1}{\zeta^\zeta(1-\zeta)^{1-\zeta}}\left[\left(\int_0^1 (X_j^N)^{\frac{\sigma-1}{\sigma}}\,\mathrm{d}j\right)^{\frac{\sigma}{\sigma-1}}\right]^\zeta \left[\exp\left(\int_0^1 \ln X_j^S\,\mathrm{d}j\right)\right]^{1-\zeta}, \tag{63}$$

$$y_j^N = A\exp\left(\int_0^{a_j} \ln x_{ij}^N\,\mathrm{d}i + \int_{a_j}^1 \ln x_{kj}^S\,\mathrm{d}k\right), \qquad j\in[0,1], \tag{64}$$

$$y_j^S = A\ell_j, \qquad n = \int_0^1 \ell_j\,\mathrm{d}j, \tag{65}$$

$$y_j^N = X_j^N + \int_0^1 x_{ji}^N \mathbf{1}\{i \text{ buys from } j\}\,\mathrm{d}i, \qquad j\in[0,1],$$

$$y_j^S = X_j^S + \int_0^1 x_{ji}^S \mathbf{1}\{i \text{ buys from } j\}\,\mathrm{d}i, \qquad j\in[0,1]. \tag{66}$$

The indicator follows the same fixed link-assignment rule as in H.1. No symmetry of quantities or customer bases has yet been imposed. The feasible set imposes $0 \le \alpha_j' < 1$ for every j.

To write the Lagrangian compactly, let $\mathcal{F}^F$ denote the right-hand side of (63) and let $\mathcal{F}_j^N$ denote the right-hand side of (64). Substituting the two production constraints into market

clearing gives

$$\begin{aligned}\mathcal{L}^{SP} = u(c,n) &+ \beta\mathbb{E}\big[V^{SP}\big(\{\alpha'_j\}, A'\big)\big|A\big] + \Lambda\left[Y - \int_0^1 \Phi(\alpha_j, \alpha'_j)\,\mathrm{d}j - c\right] \\ &+ \lambda^F\left[\mathcal{F}^F - Y\right] \\ &+ \int_0^1 \lambda_j^N\left[\mathcal{F}_j^N - X_j^N - \int_0^1 x_{ji}^N \mathbf{1}\{i \text{ buys from } j\}\,\mathrm{d}i\right]\mathrm{d}j \\ &+ \int_0^1 \lambda_j^S\left[A\ell_j - X_j^S - \int_0^1 x_{ji}^S \mathbf{1}\{i \text{ buys from } j\}\,\mathrm{d}i\right]\mathrm{d}j + \omega\left[n - \int_0^1 \ell_j\,\mathrm{d}j\right]. \end{aligned} \tag{67}$$

The multipliers are shadow values; no market price is a planner choice. Taking variations in this substituted Lagrangian with respect to its choice variables gives

$$[c]: \qquad \Lambda = u_c. \tag{68a}$$

$$[n]: \qquad \omega = -u_n. \tag{68b}$$

$$[Y]: \qquad \lambda^F = \Lambda. \tag{68c}$$

$$[\ell_j]: \qquad A\lambda_j^S = \omega. \tag{68d}$$

$$[X_j^N]: \qquad \lambda^F \zeta Y \frac{(X_j^N)^{-1/\sigma}}{\int_0^1 (X_h^N)^{\frac{\sigma-1}{\sigma}}\,\mathrm{d}h} = \lambda_j^N. \tag{68e}$$

$$[X_j^S]: \qquad \lambda_j^S = \lambda^F \frac{\partial \mathcal{F}^F}{\partial X_j^S} = \lambda^F(1-\zeta)\frac{Y}{X_j^S}. \tag{68f}$$

$$[x_{ij}^N]: \qquad \lambda_j^N \frac{y_j^N}{x_{ij}^N} = \lambda_i^N, \qquad i \in [0, a_j]. \tag{68g}$$

$$[x_{kj}^S]: \qquad \lambda_j^N \frac{y_j^N}{x_{kj}^S} = \lambda_k^S, \qquad k \in [a_j, 1]. \tag{68h}$$

$$[\alpha'_j]: \qquad \Lambda\Phi_2(\alpha_j, \alpha'_j) = \beta\mathbb{E}\left[\frac{\delta V^{SP}(\{\alpha'_h\}, A')}{\delta \alpha'_j}\middle| A\right]. \tag{68i}$$

We now impose symmetry. Write

$$\begin{aligned} &\alpha_j = \alpha, && \alpha'_j = \alpha', && && \\ &X_j^N = X^N, && X_j^S = X^S, && y_j^N = y^N, && \ell_j = n, \\ &x_{ij}^N = x^N, && x_{kj}^S = x^S, && \lambda_j^N = \lambda^N, && \lambda_j^S = \lambda^S. \end{aligned}$$

Under the same regular homogeneous link assignment maintained in H.1, each network seller serves a measure $(1-\delta)\alpha$ of network buyers and each simple seller serves a measure $1-(1-\delta)\alpha$.

First consider the network- and simple-input conditions (68g) and (68h). For a network input, symmetry makes the buyer's and seller's shadow values equal to the same λ^N. For a simple input, the seller's shadow value is λ^S. Therefore,

$$\begin{aligned} \lambda^N \frac{y^N}{x^N} = \lambda^N \quad &\Longrightarrow \quad x^N = y^N, \\ \lambda^N \frac{y^N}{x^S} = \lambda^S \quad &\Longrightarrow \quad \lambda^S x^S = \lambda^N y^N. \end{aligned} \tag{69}$$

Next use the two market-clearing conditions in (66). A network seller supplies x^N to a measure $(1-\delta)\alpha$ of buyers. Using $x^N = y^N$, its clearing condition becomes

$$\begin{aligned} y^N &= X^N + (1-\delta)\alpha x^N = X^N + (1-\delta)\alpha y^N, \\ X^N &= \left[1-(1-\delta)\alpha\right] y^N. \end{aligned} \tag{70}$$

A simple seller supplies x^S to the complementary measure of network buyers. Since its output is An, its clearing condition is

$$An = X^S + \left[1-(1-\delta)\alpha\right] x^S. \tag{71}$$

At symmetry, (68e) and (68f) give

$$\lambda^N = \lambda^F \frac{\zeta Y}{X^N}, \qquad\qquad \lambda^S = \lambda^F \frac{(1-\zeta)Y}{X^S}. \tag{72}$$

Together with the input conditions and market-clearing equations (69)– (71), these shadow values give

$$\frac{\left[1-(1-\delta)\alpha\right] x^S}{X^S} = \frac{\lambda^N X^N}{\lambda^S X^S} = \frac{\zeta}{1-\zeta}. \tag{73}$$

Simple-goods clearing then yields

$$X^S = (1-\zeta)An, \qquad [1-(1-\delta)\alpha]\, x^S = \zeta An. \tag{74}$$

Network production and the network-input condition imply

$$y^N = x^N = A^{\frac{1}{1-(1-\delta)\alpha}} x^S.$$

Hence

$$\begin{aligned} X^N &= [1-(1-\delta)\alpha]\, y^N \\ &= A^{\frac{1}{1-(1-\delta)\alpha}} [1-(1-\delta)\alpha]\, x^S \\ &= \zeta A^{1+\frac{1}{1-(1-\delta)\alpha}} n. \end{aligned} \tag{75}$$

Substituting this result and (74) into the normalized final technology gives

$$\begin{aligned} Y &= \frac{(X^N)^\zeta (X^S)^{1-\zeta}}{\zeta^\zeta (1-\zeta)^{1-\zeta}} \\ &= A^{1+\frac{\zeta}{1-(1-\delta)\alpha}} n. \end{aligned} \tag{76}$$

The elasticity σ governs curvature across network varieties, but it drops out of the symmetric planner allocation.

The intratemporal conditions in (68) imply

$$-\frac{u_n(c,n)}{u_c(c,n)} = A(1-\zeta)\frac{Y}{X^S} = A^{1+\frac{\zeta}{1-(1-\delta)\alpha}}. \tag{77}$$

Under GHH utility, at a common (α, A),

$$n^{SP}(\alpha, A) = \left[\frac{1}{\eta} A^{1+\frac{\zeta}{1-(1-\delta)\alpha}}\right]^\chi = \left(\frac{\sigma}{\sigma-1}\right)^{\zeta\chi} n^{CE}(\alpha, A). \tag{78}$$

The planner therefore removes the static retail-markup distortion.

Applying the envelope theorem to (67), a symmetric increase in α moves the input

boundary from the simple to the network sector. Its three current-production effects give

$$V_\alpha^{SP}(\alpha, A) = -\Lambda\Phi_1(\alpha, \alpha') + (1-\delta)\left[\lambda^N y^N \ln\left(\frac{x^N}{x^S}\right) - \lambda^N x^N + \lambda^S x^S\right]. \tag{79}$$

The last two terms cancel by (69). Moreover, network production implies

$$\ln\left(\frac{x^N}{x^S}\right) = \frac{\ln A}{1-(1-\delta)\alpha}.$$

Finally, equations (72) and (70) give

$$\frac{\lambda^N}{\lambda^F} y^N = \frac{\zeta Y}{1-(1-\delta)\alpha}. \tag{80}$$

Using $\Lambda = \lambda^F = u_c$, the envelope becomes

$$V_\alpha^{SP}(\alpha, A) = u_c(c, n)\left[\zeta Y \ln A \frac{1-\delta}{\left[1-(1-\delta)\alpha\right]^2} - \Phi_1(\alpha, \alpha')\right]. \tag{81}$$

The resulting Euler equation is

$$\Phi_2(\alpha, \alpha') = \mathbb{E}\left[\beta \frac{u_c(c', n')}{u_c(c, n)}\left\{\zeta Y' \ln A' \frac{1-\delta}{\left[1-(1-\delta)\alpha'\right]^2} - \Phi_1(\alpha', \alpha'')\right\}\middle| A\right]. \tag{82}$$

At a common (Y, α, A), the planner's gross return to α is $\sigma/(\sigma-1)$ times the decentralized return in (60). In the two allocations, Y and n also differ. The wedge arises because the planner removes the retail markup on network varieties; it is not a separate customer-base externality.

H.3 The adjustment cost

We use the following adjustment cost:

$$\Phi(\alpha_t, \alpha_{t+1}) = \frac{\mu}{2}\left[\ln\left(\frac{1-(1-\delta)\alpha_t}{1-\alpha_{t+1}}\right)\right]^2. \tag{83}$$

The main text provides the matching interpretation and related literature. For completeness, we record the derivation. Let U_t and V_t denote the masses on the two sides of the matching market. Given search intensity s_t, the mass of new matches is

$$M_t = V_t \left[1 - \exp\left(-s_t \frac{U_t}{V_t}\right)\right]. \tag{84}$$

We assume that producing s_t requires the convex resource cost $\mu s_t^2/2$ units of the final good. In our symmetric network economy,

$$U_t = V_t = 1 - (1-\delta)\alpha_t, \qquad M_t = \alpha_{t+1} - (1-\delta)\alpha_t.$$

Eliminating s_t from the matching technology and its resource cost gives exactly (83).[1]

References

Antràs, Pol, Davin Chor, Thibault Fally, and Russell Hillberry. 2012. "Measuring the Upstreamness of Production and Trade Flows." *American Economic Review* 102 (3):412–16. URL https://www.aeaweb.org/articles?id=10.1257/aer.102.3.412.

Dupraz, Stéphane, Emi Nakamura, and Jón Steinsson. 2025. "A plucking model of business cycles." *Journal of Monetary Economics* 152:103766.

[1] Since $U_t/V_t = 1$, the matching technology implies $M_t = V_t(1 - e^{-s_t})$. Substituting for M_t and V_t gives $1 - \alpha_{t+1} = [1 - (1-\delta)\alpha_t]e^{-s_t}$, and therefore

$$s_t = \ln\left(\frac{1 - (1-\delta)\alpha_t}{1 - \alpha_{t+1}}\right).$$

Substitution into the resource cost $\mu s_t^2/2$ gives (83). The literal matching problem describes new link formation and therefore $s_t \geq 0$. When the quantitative model permits active link destruction, we interpret $s_t < 0$ as signed reconfiguration intensity and apply the same quadratic resource cost.